Ignacio Magaña Hernandez

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Research Interests

Gravitational-wave astrophysics and cosmology: where and how binary black holes form and merge, connecting their populations to host galaxies with DESI; dark- and spectral-siren measurements of the Hubble constant; nonparametric Bayesian inference of compact-binary populations.

Employment

Carnegie Mellon University — Pittsburgh, PA
Research Physicist — June 2026 – Present

Carnegie Mellon University — Pittsburgh, PA
McWilliams Postdoctoral Fellow — June 2023 – June 2026

University of Wisconsin–Milwaukee — Milwaukee, WI
Graduate Research Fellow (NSF GRFP and UWM AOP) — September 2017 – May 2023

The Chinese University of Hong Kong — Hong Kong
Undergraduate Research Assistant — March 2016 – January 2017

Caltech — Pasadena, CA
LIGO SURF Fellow — June 2015 – September 2015

Education

University of Wisconsin–Milwaukee — Milwaukee, WI
Ph.D., Physics (advisor: Patrick R. Brady) — September 2017 – May 2023
Thesis: *Cosmology, Lensing, and Modified Gravity with Gravitational Waves*

University of California, Santa Barbara — Santa Barbara, CA
B.S., Physics (Highest Honors) — September 2014 – September 2017

Foothill College — Los Altos Hills, CA
Transfer Student, Physics — September 2012 – June 2014

Awards & Fellowships

- **CMU McWilliams Postdoctoral Fellowship** — 2023 – 2026
  McWilliams Center for Cosmology and Astrophysics two-year postdoctoral fellowship, renewed for a third year.
- **UWM Advanced Opportunity Program Fellowship** — 2020 – 2023
  UWM fellowship supporting underrepresented graduate students (three years of support).
- **NSF Graduate Research Fellowship** — 2017 – 2022
  Five-year NSF fellowship (three years of stipend support).
- **NASA Wisconsin Space Grant Consortium Fellowship** — 2019 – 2020, 2021 – 2022
  NASA-funded fellowship for graduate research in STEM at Wisconsin institutions.
- **UWM Physics Department David Lichtman Memorial Fellowship** — Spring 2020
  Departmental fellowship awarded for exceptional academic achievement in physics.
- **Victor M. Blanco Undergraduate Research Fellowship (LIGO SURF)** — Summer 2015
  Summer Research Fellowship at Caltech/LIGO, supporting undergraduates in GW science.
- **NSF S-STEM Scholarship** — Fall 2013 – Spring 2014
  NSF Scholarships in STEM program providing support to students pursuing STEM degrees.

Publication Summary

	Papers	Citations	h-index
First/second-author papers, excluding student-led	14	785	12
Excluding collaboration papers	39	2.6k	20
Including LIGO–Virgo collaboration papers	134	64k	78

Citations according to inspirehep.net in October 2026; the first two rows count the papers listed below, except papers in preparation.

First/Second Author Publications

1. *A Quiet Night in the Millihertz Sky? LISA Forecasts from LIGO/Virgo/KAGRA Populations*
   I. Magaña Hernandez and A. Palmese
   (arXiv:2610.37617, 0 citations) *Submitted to ApJ*
2. *Lucky Strikes: On the Origins of GW190814 Through Isolated Binary Evolution*
   I. Magaña Hernandez and K. Breivik
   (arXiv:2511.16648, 0 citations) *Submitted to ApJL*
3. *Spectral siren cosmology from gravitational-wave observations in GWTC-4.0*
   I. Magaña Hernandez and A. Palmese
   (arXiv:2509.03607, 22 citations) *Submitted to Phys. Rev. Lett.*
4. *Astrophysics informed Gaussian processes for gravitational-wave populations: Evidence for the onset of the pair-instability supernova mass gap*
   I. Magaña Hernandez and A. Palmese
   (arXiv:2508.19208, 18 citations) *Submitted to Phys. Rev. Lett.*
5. *Evidence of a new feature in the binary black hole mass distribution at 70 $M_\odot$ from gravitational-wave observations*
   I. Magaña Hernandez and A. Palmese
   Phys. Rev. D 111, 083031 (2025) (arXiv:2407.02460, 22 citations)
6. *Searching for binary black hole subpopulations in gravitational-wave data using binned Gaussian processes*
   A. Ray, **I. Magaña Hernandez**, K. Breivik and J. D. E. Creighton
   ApJ 991, 17 (2025) (arXiv:2404.03166, 53 citations)
7. *Beyond Gaps and Bumps: Spectral Siren Cosmology with Non-Parametric Population Models*
   I. Magaña Hernandez and A. Ray
   (arXiv:2404.02522, 34 citations) *Submitted to Phys. Rev. D*
8. *On the association of GW190425 with its potential electromagnetic counterpart FRB 20190425A*
   I. Magaña Hernandez, V. D'Emilio, S. Morisaki, M. Bhardwaj and A. Palmese
   ApJL 971, L5 (2024) (arXiv:2404.02496, 14 citations)
9. *Nonparametric inference of the population of compact binaries from gravitational-wave observations using binned Gaussian processes*
   A. Ray, **I. Magaña Hernandez**, S. Mohite, J. D. E. Creighton and S. Kapadia
   ApJ 957, 37 (2023) (arXiv:2304.08046, 95 citations)
10. *Measuring the polarization content of gravitational waves with strongly lensed binary black hole mergers*
    I. Magaña Hernandez
    (arXiv:2211.01272, 12 citations) *Submitted to Phys. Rev. D*
11. *Constraining the number of spacetime dimensions from GWTC-3 binary black hole mergers*
    I. Magaña Hernandez
    Phys. Rev. D 107, 084033 (2023) (arXiv:2112.07650, 23 citations)
12. *Bayesian statistical framework for identifying strongly lensed gravitational-wave signals*
    R. K. L. Lo and **I. Magaña Hernandez**
    Phys. Rev. D 107, 123015 (2023) (arXiv:2104.09339, 103 citations)
13. *Identifying strong gravitational-wave lensing during the second observing run of Advanced LIGO and Advanced Virgo*
    X. Liu, **I. Magaña Hernandez** and J. D. E. Creighton
    ApJ 908, 97 (2021) (arXiv:2009.06539, 109 citations)
14. *Cosmological inference using gravitational wave standard sirens: A mock data analysis*
    R. Gray, **I. Magaña Hernandez** et al.
    Phys. Rev. D 101, 122001 (2020) (arXiv:1908.06050, 280 citations)

Student-Led Publications

Papers led by students I mentor (see Student Mentoring).

1. *Flexible Proposal Distributions for Gravitational-Wave Population Inference with Gaussian Mixture Models*
   S. M. C. Yeung, **I. Magaña Hernandez** and J. D. E. Creighton
   (arXiv:2610.30187, 0 citations) *Submitted to Phys. Rev. D*
2. *Hostbusters: constraining H_0 and binary formation channels through gravitational-wave host galaxy associations*
   M. L. Cross-Parkin, **I. Magaña Hernandez**, A. Palmese, P. D. Lasky and E. Thrane
   (arXiv:2610.25866, 0 citations) *Submitted to ApJ*

3. *Normalizing Flow Emulator for Gravitational Wave Detection Probability in the LIGO/Virgo/KAGRA O4 Observing Run*
   J. Gassert, **I. Magaña Hernandez**, A. J. Amsellem, A. Palmese and B. Póczos
   (arXiv:2610.00306, 0 citations) *Submitted to Phys. Rev. D*
4. *One Year and One Night to 1% in H_0: Efficient Spectroscopic Strategy for Dark Siren Cosmology*
   Y. Dang, A. J. Amsellem, **I. Magaña Hernandez**, A. Palmese and B. S. Sathyaprakash
   (arXiv:2607.17284, 1 citation) *Submitted to Phys. Rev. Lett.*
5. *Data-Driven Trends and Subpopulations in the Gravitational Wave Binary Black Hole Merger Population with UMAP*
   A. J. Amsellem, **I. Magaña Hernandez**, A. Palmese and J. Gassert
   (arXiv:2603.06566, 2 citations) *Submitted to Phys. Rev. X*
6. *Probing the environment around GW170817 with DESI: insights on galaxy group peculiar velocities for standard siren measurements*
   A. J. Amsellem et al. (including **I. Magaña Hernandez**)
   ApJ 1001, 157 (2026) (arXiv:2512.08818, 6 citations)

Other Publications

1. *Electromagnetic Follow-up of the Sub-Solar Mass Gravitational Wave Candidate S251112cm: Kilonova Constraints and a Coincident IIb Supernova*
   X. J. Hall et al. (including **I. Magaña Hernandez**)
   (arXiv:2605.10940, 11 citations) *Submitted to ApJL*
2. *Wide Jets or Low Rates: Reconciling Short GRB and Gravitational-Wave Neutron Star Merger Rates*
   K. Kunnumkai, A. Palmese, B. O'Connor, A. Farah and **I. Magaña Hernandez**
   (arXiv:2604.05046, 7 citations) *Submitted to ApJ*
3. *AT2025ulz and S250818k: Investigating early time observations of a subsolar mass gravitational-wave binary neutron star merger candidate*
   X. J. Hall et al. (including **I. Magaña Hernandez**)
   (arXiv:2510.24620, 19 citations) *Submitted to ApJL*
4. *AT2025ulz and S250818k: Deep X-ray and radio limits on off-axis afterglow emission and prospects for future discovery*
   B. O'Connor et al. (including **I. Magaña Hernandez**)
   ApJL 995, L47 (2025) (arXiv:2510.23728, 14 citations)
5. *AT2025ulz and S250818k: Leveraging DESI spectroscopy in the hunt for a kilonova associated with a subsolar-mass gravitational-wave candidate*
   X. J. Hall et al. (including **I. Magaña Hernandez**)
   ApJL 1001, L20 (2026) (arXiv:2510.23723, 15 citations)
6. *Kilonova Constraints for the LIGO/Virgo/KAGRA Neutron Star Merger Candidate S250206dm: GW-MMADS Observations*
   L. Hu et al. (including **I. Magaña Hernandez**)
   ApJL 990, L46 (2025) (arXiv:2506.22626, 17 citations)
7. *A Dark Siren Measurement of the Hubble Constant with the LIGO/Virgo Gravitational Wave Event GW190412 and DESI Galaxies*
   W. Ballard, A. Palmese, **I. Magaña Hernandez** et al.
   Res. Notes AAS 7, 250 (2023) (arXiv:2311.13062, 35 citations)
8. *Follow-up analyses to the O3 LIGO–Virgo–KAGRA lensing searches*
   J. Janquart et al. (including **I. Magaña Hernandez**)
   MNRAS 526, 3832 (2023) (arXiv:2306.03827, 87 citations)
9. *Challenges for Fast Radio Bursts as Multimessenger Sources from Binary Neutron Star Mergers*
   M. Bhardwaj, A. Palmese, **I. Magaña Hernandez**, V. D'Emilio and S. Morisaki
   ApJ 977, 122 (2024) (arXiv:2306.00948, 31 citations)
10. *Localization of binary neutron star mergers with a single Cosmic Explorer*
    P. Baral, S. Morisaki, **I. Magaña Hernandez** and J. D. E. Creighton
    Phys. Rev. D 108, 043010 (2023) (arXiv:2304.09889, 19 citations)
11. *The Hitchhiker's guide to the galaxy catalog approach for dark siren gravitational-wave cosmology*
    J. Gair et al. (including **I. Magaña Hernandez**)
    AJ 166, 22 (2023) (arXiv:2212.08694, 122 citations)

12. *A joint ranking statistic for multi-messenger astronomical searches with gravitational waves*
B. Piotrzkowski, A. Baylor and **I. Magaña Hernandez**
Class. Quantum Grav. 39, 085010 (2022) (arXiv:2111.12814, 8 citations)
13. *Rapid model comparison of equations of state from gravitational wave observation of binary neutron star coalescences*
S. Ghosh, X. Liu, J. D. E. Creighton, W. Kastaun, G. Pratten and **I. Magaña Hernandez**
Phys. Rev. D 104, 083003 (2021) (arXiv:2104.08681, 25 citations)
14. *Gravitational wave friction in light of GW170817 and GW190521*
S. Mastrogiovanni, L. Haegel, C. Karathanasis, **I. Magaña Hernandez** and D. A. Steer
JCAP 02 (2021) 043 (arXiv:2010.04047, 45 citations)
15. *First measurement of the Hubble parameter from bright binary black hole GW190521*
S. Mukherjee et al. (including **I. Magaña Hernandez**)
(arXiv:2009.14199, 73 citations) *Preprint*
16. *Current observations are insufficient to confidently associate the binary black hole merger GW190521 with AGN J124942.3 + 344929*
G. Ashton, K. Ackley, **I. Magaña Hernandez** et al.
Class. Quantum Grav. 38, 235004 (2021) (arXiv:2009.12346, 114 citations)
17. *A statistical standard siren measurement of the Hubble Constant from the LIGO/Virgo gravitational wave compact object merger GW190814 and Dark Energy Survey galaxies*
A. Palmese et al. (including **I. Magaña Hernandez**)
ApJL 900, L33 (2020) (arXiv:2006.14961, 177 citations)
18. *Bayesian inference for compact binary coalescences with bilby: Validation and application to the first LIGO–Virgo gravitational-wave transient catalogue*
I. M. Romero-Shaw et al. (including **I. Magaña Hernandez**)
MNRAS 499, 3295 (2020) (arXiv:2006.00714, 649 citations)
19. *A standard siren measurement of the Hubble Constant from GW170817 without the electromagnetic counterpart*
M. Fishbach, R. Gray, **I. Magaña Hernandez** et al.
ApJL 871, L13 (2019) (arXiv:1807.05667, 308 citations)

LIGO–Virgo Collaboration Papers with Major Personal Contributions

1. *GW190814: Gravitational Waves from the Coalescence of a 23 Solar Mass Black Hole with a 2.6 Solar Mass Compact Object*
LIGO Scientific and Virgo Collaborations (including **I. Magaña Hernandez**)
ApJL 896, L44 (2020) (arXiv:2006.12611, 2128 citations) **Led and wrote the Hubble constant analysis**
2. *A Gravitational-wave Measurement of the Hubble Constant Following the Second Observing Run of Advanced LIGO and Virgo*
LIGO Scientific and Virgo Collaborations (including **I. Magaña Hernandez**)
ApJ 909, 218 (2021) (arXiv:1908.06060, 425 citations) **Paper writing and analysis team member**
3. *First measurement of the Hubble constant from a dark standard siren using the Dark Energy Survey galaxies and the LIGO/Virgo binary-black-hole merger GW170814*
M. Soares-Santos, A. Palmese, et al. (DES, LIGO Scientific and Virgo Collaborations, including **I. Magaña Hernandez**)
ApJL 876, L7 (2019) (arXiv:1901.01540, 381 citations) **Cross-checked the H_0 result with `gwcosmo`**

Co-author of 92 further LIGO–Virgo collaboration papers, not listed (included in the Publication Summary).

Papers in Preparation

1. *Spectroscopic dark sirens: the Hubble constant from the binary black holes of GWTC-5 and DESI*
I. Magaña Hernandez, A. Palmese et al. (DESI Collaboration paper) *In preparation*
2. *A common effective-spin mixture across mass, mass ratio, and redshift in GWTC-5*
I. Magaña Hernandez and A. Palmese *In preparation*
3. *Multitracer dark sirens: joint inference of the Hubble constant and the AGN-hosted merger fraction*
I. Magaña Hernandez, K. Storey-Fisher and A. Palmese *In preparation*
4. *Hostbusters II: how binary black hole masses and spins depend on their host galaxies*
I. Magaña Hernandez et al. *In preparation*

Papers in preparation are not counted in the Publication Summary.

Student Mentoring

Madeline Cross-Parkin — University of Queensland
PhD student (advisor: T. M. Davis) — June 2025 – Present
Host-galaxy associations and H_0 (*Hostbusters*).

Ariel Amsellem — Carnegie Mellon University
PhD student (advisor: A. Palmese) — September 2023 – Present
Standard siren cosmology (2025 Lederman Fellow).

Julius Gassert — LMU Munich (visiting CMU)
PhD student (advisor: D. Gruen) — August 2025 – Present
Machine learning for GW selection effects.

Yixuan (Wendy) Dang — Pennsylvania State University
PhD student (advisor: B. S. Sathyaprakash) — August 2025 – Present
Spectroscopic strategies for dark sirens.

Simon Yeung — University of Wisconsin–Milwaukee
PhD student (advisor: J. D. E. Creighton) — September 2023 – Present
GW population systematics.

Julia Yoon — Carnegie Mellon University
Undergraduate student — September 2024 – September 2026
Dark siren cosmology with GW170817.

Kaitlyn Pak — Carnegie Mellon University
Undergraduate student — September 2025 – June 2026
Multi-messenger associations with GWs.

Diego Michel — Carnegie Mellon University
Undergraduate student — September 2023 – May 2024
Gaussian processes for GW populations.

Grants & Computing Allocations

- **NSF ACCESS Allocation PHY240308 (PI)** — 2024 – Present
  Computing allocation for GPU-accelerated gravitational-wave population and cosmology inference: 400K service units (Explore) and 1.5M service units (Discover).
- **CMU McWilliams Seed Grant** — 2025
  Internal grant funding exploratory projects to generate preliminary results for high-impact research.

Service

- **Referee:** Physical Review D, Physical Review Letters, The Astrophysical Journal (ApJ/ApJL), Monthly Notices of the Royal Astronomical Society, Nature Astronomy.
- **Session chair:** Parallel session 8a "GW/MMA II: theory", COSMO-25, Pittsburgh (2025).
- **Journal club organizer:** Carnegie Mellon University; University of Wisconsin–Milwaukee.
- **DESI:** lead of the dark-siren analysis; lead author of its key paper (H_0 from GWTC-5 and DESI; in prep.).
- **LIGO Scientific Collaboration:** led and wrote the Hubble-constant analysis of GW190814; paper-writing and analysis team for the O2 Hubble-constant paper; co-developer of `gwcosmo`.

Invited Seminars

1. **American Museum of Natural History** — October 2025
   Binary black hole populations and cosmology with gravitational-wave observations and DESI
2. **Vanderbilt University** — April 2025
   Binary black hole populations and cosmology with gravitational-wave observations and DESI
3. **KAGRA Seminar (Gravitational-Wave Research Exchange Meeting)** — April 2025
   Binary black hole populations and cosmology with gravitational-wave observations
4. **Stanford University (KIPAC Seminar)** — March 2025
   Binary black hole populations and cosmology with gravitational-wave observations
5. **Carnegie Mellon University (STAMPS Seminar)** — January 2025
   Binary black hole populations and cosmology
6. **Montclair State University** — September 2023
   Cosmology, Lensing, and Modified Gravity with Gravitational Waves

7. **Carnegie Mellon University** September 2023
Testing general relativity with strongly lensed gravitational waves
8. **Cardiff University** September 2022
Cosmology and modified gravity with binary black holes
9. **UC Berkeley** June 2022
Constraining the number of spacetime dimensions from GWTC-3 binary black hole mergers
10. **University of Wisconsin–Milwaukee** April 2022
Cosmology and modified gravity with binary black hole mergers
11. **Eötvös Loránd University** September 2019
Gravitational wave cosmology with large galaxy surveys
12. **University of Glasgow** August 2019
Measuring the Hubble constant with gravitational-wave standard sirens
13. **The Chinese University of Hong Kong** February 2019
A measurement of the Hubble constant from the second observing run of Advanced LIGO–Virgo
14. **Nagoya University** February 2019
A measurement of the Hubble constant from the second observing run of Advanced LIGO–Virgo

Contributed Talks

1. **Gravitational Wave Physics and Astronomy Workshop** December 2025
Dark Siren Cosmology with the Dark Energy Spectroscopic Instrument (DESI)
2. **COSMO-25** October 2025
Gravitational-wave dark siren cosmology with DESI
3. **Multimessenger Astronomy in the Era of Foundational AI** August 2025
Trends in the GWTC-3 Binary Black Hole Merger Population with UMAP
4. **DESI Summer Collaboration Meeting** July 2025
Gravitational-wave cosmology with DESI
5. **APS Global Physics Summit** March 2025
Lucky Strikes: Unraveling the asymmetric origins of GW190814 through binary field evolution
6. **The 6th Neighborhood Workshop** April 2024
Gravitational wave populations and cosmology
7. **Gravitational-wave populations: what's next?** July 2023
Non-parametric inference of the population of compact binaries from gravitational wave observations using binned Gaussian processes
8. **APS April Meeting** April 2023
Measuring the polarization content of gravitational waves with strongly lensed binary black hole mergers
9. **11th Australasian Conference on General Relativity and Gravitation** (virtual) February 2022
Constraining the number of spacetime dimensions from GWTC-3 binary black hole mergers
10. **Midwest Relativity Meeting** November 2021
Constraining the number of spacetime dimensions from GWTC-2 binary black hole mergers
11. **Midwest Relativity Meeting** October 2020
On the candidate electromagnetic counterpart of GW190521 and its implications on the Hubble constant
12. **Gravitational Wave Physics and Astronomy Workshop** October 2019
A gravitational-wave measurement of the Hubble constant following the second observing run of Advanced LIGO and Virgo
13. **Multi-Messenger Astrophysics in the Gravitational Wave Era** September 2019
A gravitational-wave measurement of the Hubble constant following the second observing run of Advanced LIGO and Virgo
14. **YITP Asian-Pacific Winter School and Workshop on Gravitation and Cosmology** February 2019
A standard siren measurement of the Hubble constant from GW170817 without the electromagnetic counterpart
15. **Midwest Relativity Meeting** October 2018
Cosmological inference using gravitational wave standard sirens: A mock data challenge

Other Workshops

1. **Multi-messenger Transients from Binary Mergers and Stellar Explosions** August 2024
   Aspen Center for Physics
2. **YKIS2019: Black Holes and Neutron Stars with Gravitational Waves** October 2019
   Yukawa Institute for Theoretical Physics
3. **The New Era of Gravitational-Wave Physics and Astrophysics** May – July 2019
   Kavli Institute for Theoretical Physics
4. **Les Houches Summer School: Gravitational Waves** July 2018
   École de Physique des Houches
5. **Summer School on Gravitational-Wave Astronomy** July – August 2016
   ICTS-TIFR

Societies/Affiliations

- Dark Energy Spectroscopic Instrument (DESI) member (June 2023 – Present)
- LIGO Scientific Collaboration member (June 2015 – June 2023; rejoining via the CMU LSC group, January 2027)
- American Physical Society
- Society for Advancement of Chicanos/Hispanics and Native Americans in Science (SACNAS)
- National Society of Hispanic Physicists

Skills

- **Statistical methods:** hierarchical Bayesian population inference with selection effects; joint population and cosmology inference (spectral and dark sirens); Gaussian processes and nonparametric population models; Bayesian model selection and parameter estimation; MCMC (including HMC/NUTS) and nested sampling.
- **Machine learning:** normalizing-flow emulators, Gaussian mixture models, dimensionality reduction and clustering (UMAP).
- **Astrophysical modeling:** binary population synthesis (COSMIC), including backward population synthesis.
- **Programming:** Python (NumPy, SciPy, Pandas, Scikit-learn), JAX and NumPyro (GPU), C, Mathematica, LaTeX.
- **Computing:** high-performance computing (SLURM clusters); GPU-accelerated inference; PI of an NSF ACCESS allocation.
- **Gravitational-wave software:** Bilby, LALSuite, PyCBC, ligo.skymap, gwcosmo (co-developer), hanabi.
- **Survey data:** DESI spectroscopic data products and galaxy catalogs; astropy, healpy/HEALPix sky maps.
- **Languages:** English (fluent), Spanish (fluent), and Korean (intermediate).

Software

- **darksirens** (designer and developer): GPU-accelerated (JAX) hierarchical population and cosmological inference for gravitational-wave sirens (github.com/ignaciomagana/darksirens-core).
- **gwcat**: preprocessing of GWTC parameter-estimation files and LVK injection sets for **darksirens** (github.com/ignaciomagana/gwcat).
- **tinyns**: lightweight JAX-based nested sampler (github.com/ignaciomagana/tinyns).
- **gwcosmo** (co-developer): LVK gravitational-wave cosmology code (git.ligo.org/lscsoft/gwcosmo).

References

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